REMEMBERING WHERE TO LOOK: CONTINUAL HIERARCHICAL EVIDENCE SELECTION FOR WHOLE-SLIDE IMAGES

Hsi-Ren Hung *[Second Author]* *Huei-Fang Yang*[*]

Department of Computer Science and Engineering, National Sun Yat-sen University, Taiwan

ABSTRACT

Whole-slide image (WSI) diagnosis is an evidence-acquisition problem: a model must select a few patches from thousands, and its prediction is only as reliable as the evidence it chooses. Continual-learning (CL) methods usually preserve classifiers or representations, but a budgeted diagnostic system can fail in a different way: its diagnostic pathway may remain intact while its selector gradually forgets *where to look*. We introduce **PathSelect**, which makes the evidence-selection policy itself the object of continual learning. To isolate this failure, the image encoder, text encoder, and cosine diagnostic head are frozen; only a hierarchical group-to-patch selector is learned. Within a task, three supervision signals answer complementary questions: whether the selected evidence predicts correctly, whether it is semantically discriminative, and which candidate patch would counterfactually reduce diagnostic loss. Across tasks, per-task LoRA residuals are merged exactly into one shared selector, while a feature-free Selection Memory preserves past behavior through dual-level distillation, a utility-preservation hinge, and replay. The key distinction is between *behavioral continuity*—how the selector scores evidence—and *functional continuity*—whether newly selected evidence remains diagnostically useful. On four sequential TCGA tasks, PathSelect raises final class-incremental accuracy by 34.6 points over sequential fine-tuning in every seed, halves forgetting relative to replay alone (0.0539 vs. 0.1102), and retains two orders of magnitude more of its originally selected evidence; under QPMIL-VL's reverse-order splits and evaluation protocol it reaches 0.854 ACC, within 0.5 points of QPMIL-VL's reported 0.859, while adapting 16.4k selector parameters per task (QPMIL-VL reports 0.365M) and storing no features, with inference on one task-free shared selector.

***Index Terms*—** whole-slide images, continual learning, evidence selection, computational pathology, LoRA

1. INTRODUCTION

A gigapixel WSI can contain tens of thousands of patches, yet a diagnosis often depends on a small subset. Practical systems therefore operate under an *evidence budget*: choose B patches, then decide. This creates a learning problem that is easy to miss in conventional continual learning. A model may preserve its classifier and representation reasonably well, yet after learning new disease cohorts it can still fail because it no longer chooses the tissue that once supported the correct decision.

We therefore ask a narrower question:

> *When diagnostic tasks arrive sequentially, does a model forget where to look, and can that ability be preserved, and its loss measured, directly?*

PathSelect is designed around this question rather than around a collection of modules. Its first principle is **attribution by freezing**. CONCH image/text encoders [1] and the cosine diagnostic rule are fixed throughout training. There is no trainable classifier. The only learned object is an evidence-selection policy. For paired methods evaluated at the same checkpoint and class set the diagnostic pathway is identical, so between-method differences arise from selector updates; within-method class-incremental changes over time additionally include label-space expansion. Direct selection metrics provide an additional head-independent measurement.

Our hierarchical substrate follows the coarse-to-fine tissue-aware idea of HistoSelect [2]: first allocate attention among tissue groups, then select patches within each group. We do *not* claim this hierarchy itself as our primary novelty. Instead, PathSelect starts where such selectors normally stop: after a selector has learned where to look, how can one keep that ability when tasks arrive sequentially?

This perspective leads to a deliberate separation between two kinds of learning. Within a task, PathSelect must *learn where to look now*. Across tasks, it must *keep looking well*. The first is trained by diagnostic, semantic, and counterfactual-utility supervision. The second is trained by three preservation questions: (i) do I still score old evidence similarly, (ii) even if I select differently, is the new evidence at least as useful, and (iii) can I still solve the old diagnostic problem?

Contributions.

- We formulate continual WSI diagnosis with the *evidence-selection policy*—rather than a classifier—as the learned, preserved, and measured object, isolated by a frozen diagnostic pathway; to our knowledge, prior WSI continual-learning methods adapt MIL, attention, or localization components, whereas recent budgeted evidence-selection and active-perception methods are not continual; PathSelect isolates the intersection—the diagnostic pathway is fixed, and a hard, budgeted evidence selector is the only learned and directly evaluated continual object.
- We introduce a fixed-cost continual protocol in which per-task LoRA residuals are merged into one shared hierarchical selector, while a feature-free Selection Memory preserves both *selection behavior* and *selection usefulness*; under QPMIL-VL's splits and metrics it reaches 0.854 ACC, within 0.5 points of the strongest published result in that comparison (0.859), while adapting 16.4k selector parameters per task against QPMIL-VL's reported 0.365M (22.3× fewer; different adapted components, reported as adaptation-cost context).

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- We pair the preservation objectives with selection-level measurements: Jaccard for identity drift and utility retention for functional drift, and show experimentally that these two notions are related but not interchangeable.

2. RELATED WORK

Evidence selection in WSIs. MIL methods such as ABMIL, CLAM, and TransMIL [3, 4, 5] aggregate patch features with learned attention, but selection is entangled with a trained diagnostic model. HistoSelect [2] makes selection explicitly hierarchical, using tissue-aware coarse-to-fine reasoning. PathSelect inherits this group→patch inductive bias but changes the scientific setting: the diagnostic pathway is frozen and the selector is learned under a sequential task stream. HistoSelect conditions selection on a per-question query for VLM reasoning; PathSelect instead fixes a task-free tissue vocabulary, because a policy that must persist across tasks needs a grouping that does not move.

LoRA-based continual learning. Low-rank adaptation [6] provides parameter-efficient task updates. Prior CL methods use low-rank or modular updates to preserve classifier/backbone knowledge or retrieve task-specific components [7, 8, 9, 10]. PathSelect instead applies continual learning only to the evidence selector. Importantly, LoRA merging is used as a *deployment substrate*, not claimed as the preservation mechanism: our experiments show that merging alone does not prevent forgetting.

Continual WSI learning. ConSlide [11] introduced rehearsal-based continual WSI analysis on a four-organ TCGA sequence; MICIL [12] combined MIL with class-incremental learning; QPMIL-VL [13] brought vision–language prototypes to incremental WSI classification on CONCH features; and, closest to us, Li et al. [14] located forgetting in the attention layers of MIL models and proposed attention distillation with a pseudo-bag memory of informative patches. CoMEL [15] adapts MIL aggregators continually with orthogonal weighted low-rank adaptation, and MergeSlide [16] merges per-task fine-tuned models by orthogonal projection with task-to-class prompt inference. All of these retain a trainable MIL diagnostic model in which attention or aggregation is part of the learned diagnostic function. PathSelect differs on four axes: the entire diagnostic pathway is frozen and the selector is the sole learned object; selection is a hard, budgeted set rather than soft attention weights; preservation targets the utility of the selected set in addition to score behavior, so identity may change without functional regression; and selection identity and utility are measured directly rather than inferred from accuracy. Conversely, budgeted evidence selection [2] and active-perception navigation over WSIs [17] learn where to look but are not continual; PathSelect sits at the intersection of the two lines.

3. METHOD

3.1. Study object: a frozen diagnostic pathway

Tasks $\tau = 1, \dots, T$ arrive sequentially. For a slide, frozen CONCH features are

$$\{x_i\}_{i=1}^{N}, \qquad x_i \in \mathbb{R}^{512}.$$

The selector must choose an evidence set

$$\mathcal{P} = \{x_{i_1}, \dots, x_{i_B}\}, \qquad |\mathcal{P}| = B = 8.$$

Only this evidence reaches the diagnostic rule; the budget $B{=}8$ is the operating point at which learned selection peaked over random selection in a preliminary sweep (Sec. 4). If selector scores for the chosen patches are s_i, the WSI representation is the ℓ_2-normalized score-weighted mean $z = \sum_{i\in\mathcal{P}} \mathrm{softmax}(s)_i\, x_i / \|\cdot\|_2$. Prediction is the cosine similarity between z and frozen class-text embeddings $\{t_c\}$, followed by softmax. Training is task-aware; inference is task-free over all seen classes. No task identity, adapter bank, memory lookup, or trained diagnostic head is used at test time.

3.2. Hierarchical budgeted selection

Following the tissue-aware coarse-to-fine principle of HistoSelect [2], patches are assigned semantically rather than clustered. Eight fixed tissue prompts are encoded once with the frozen text encoder. Each patch is assigned by

$$G_j = \left\{ x_i : j = \arg\max_k \cos(x_i, p_k) \right\},$$

and each group prototype is

$$g_j = \mathrm{mean}(x_i \in G_j).$$

Two small two-layer MLPs (F_g and F_p, identical in structure and independent in weights) implement the policy; their input is zero-padded to a common width so that every ablation arm shares one architecture and parameter count. The group selector F_g produces group scores r_j. A largest-remainder allocation converts $\mathrm{softmax}(r)$ into integer quotas

$$b_j \in \mathbb{N}, \qquad \sum_j b_j = B.$$

The patch selector F_p produces patch scores s_i, and the top b_j patches are selected within each group. Thus the hierarchy answers two different questions: *which tissue groups deserve the budget?* and *which patches within those groups deserve the slots?*

3.3. Learning where to look: three within-task supervision signals

The within-task objective is

$$\mathcal{L}_{\mathrm{ev}} = \mathcal{L}_{\mathrm{diag}} + \beta_s \mathcal{L}_{\mathrm{sem}} + \beta_u \mathcal{L}_{\mathrm{util}}. \tag{1}$$

These terms are not redundant; they provide supervision at different levels.

Outcome supervision: $\mathcal{L}_{\mathrm{diag}}$. Cross-entropy on the frozen diagnostic logits asks: *did the final evidence set answer correctly?* It supervises the joint outcome but provides weak credit assignment among many candidate patches.

Semantic supervision: $\mathcal{L}_{\mathrm{sem}}$. A frozen class-text prior scores how diagnostically discriminative a patch appears in the CONCH embedding space. (At inference the selector receives no class text, task identity, or label; during training, class-derived information enters only as supervision targets—this prior and $\mathcal{L}_{\mathrm{diag}}$. The attribution claim concerns the frozen non-selector parameters, not the absence of class-derived supervision.) The selector distribution is softly aligned to this prior with KL divergence. This is a weak semantic anchor, not the final diagnostic target.

Counterfactual credit: $\mathcal{L}_{\mathrm{util}}$. For candidate patch i, define a marginal counterfactual gain

$$u_i = \mathcal{L}_{\mathrm{diag}}(E) - \mathcal{L}_{\mathrm{diag}}(E \cup \{x_i\}), \tag{2}$$

where E denotes the current evidence context. Positive u_i means that adding the candidate would reduce diagnostic loss; negative u_i

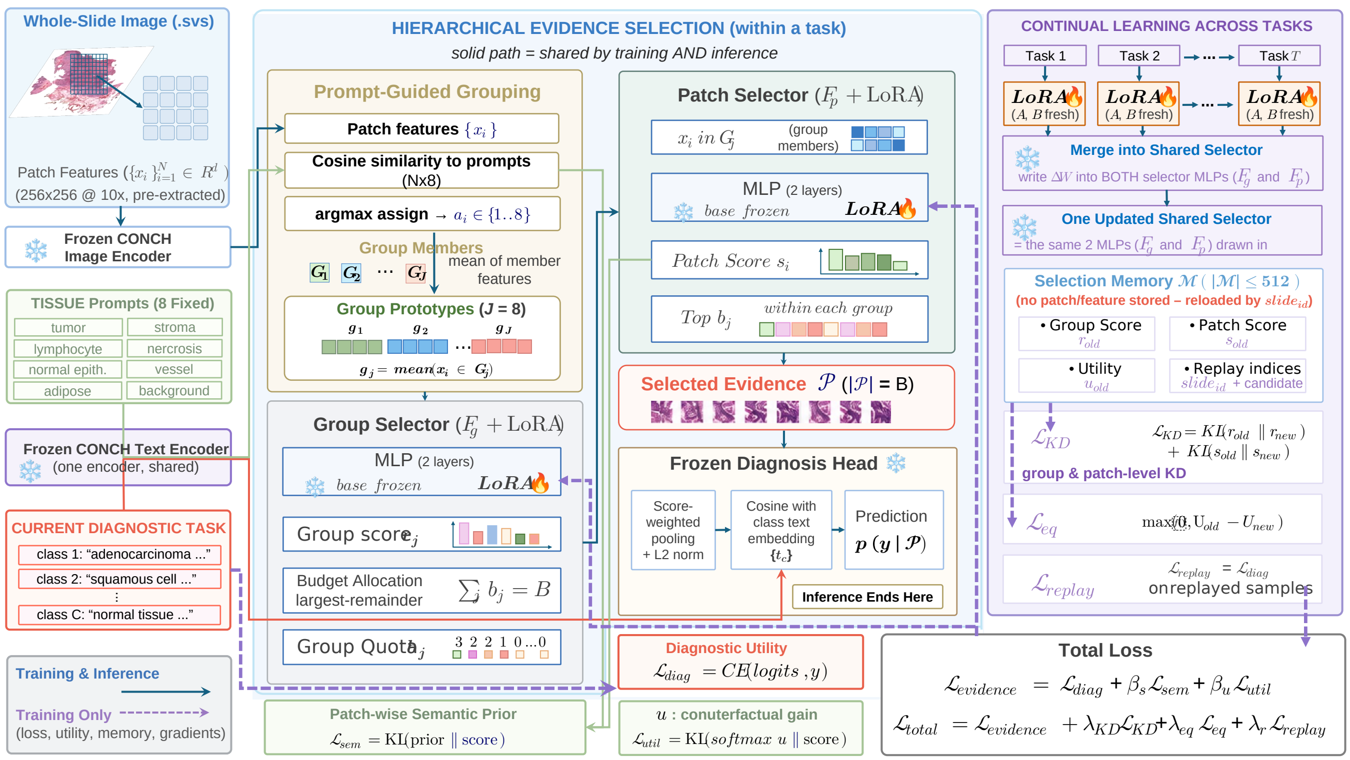

Fig. 1. **PathSelect separates learning to look from keeping looking well.** Within a task, the hierarchical selector learns a budgeted evidence policy under a frozen diagnostic pathway. Across tasks, only selector LoRA parameters are trained; task residuals are folded into one shared selector. A feature-free Selection Memory is used only during training to preserve behavior and function.

means it would be harmful. The gains are treated as a no-gradient teacher distribution, and $\mathcal{L}_{\text{util}} = \text{KL}(\text{softmax}(u) \,\|\, \text{softmax}(s))$ trains the patch selector to rank candidates by measured usefulness. Lowercase u_i is therefore a *candidate-level* teaching signal.

The value of the *whole selected evidence set* is a different quantity:

$$U(\mathcal{P}) = \log C - \text{CE}(\mathcal{P}, y). \tag{3}$$

It asks how much the complete selected set improves over an empty/uniform diagnostic state. This distinction—candidate utility u_i versus set utility U—becomes central in continual learning.

3.4. One shared selector: LoRA consolidation

For task τ, low-rank LoRA adapters are attached to the four linear layers of F_g and F_p. The base selector is frozen during the task and only the current adapters train. At the task boundary, $W_\tau = W_{\tau-1} + B_\tau A_\tau$, which is an exact rewrite of the effective weight used immediately before merging. The adapter is then reset for the next task. Thus parameter count does not grow with T and inference always uses one shared selector.

Crucially, **merging is not the anti-forgetting mechanism**. The merge-only arm performs no better systematically than naive sequential fine-tuning. LoRA supplies a clean, fixed-cost way to accumulate task updates; preservation requires additional constraints.

3.5. Keeping looking well: preserve behavior and function

After a task is learned, PathSelect stores a bounded Selection Memory $\mathcal{M}$ of at most 512 behavioral snapshots using reservoir sampling. A snapshot stores an internal sample reference, the task tag, group scores r_{old} (8 values), candidate indices (≤256), patch scores s_{old} (≤256), and utilities (≤256); it contains no raw image or feature tensor. The 512-entry memory occupies 3.11 MB when serialized. Replay nevertheless assumes authorized training-time access to the corresponding pre-extracted feature records, whose storage (19.5 GB for this benchmark) is not included in that figure.

For each current sample, one old snapshot is replayed. The continual objective is

$$\begin{aligned}\mathcal{L}_{\text{CL}} = {} & \lambda_{\text{kd}}\left[\text{KL}(r_{\text{old}} \| r_{\text{new}}) + \text{KL}(s_{\text{old}} \| s_{\text{new}})\right] \\ & + \lambda_{\text{eq}} \max(0, U_{\text{old}} - U_{\text{new}}) + \lambda_r \mathcal{L}^{\text{rep}}_{\text{diag}},\end{aligned} \tag{4}$$

with all three λ's set to 1.

Equation 4 deliberately separates three preservation questions.

Behavioral continuity: $\mathcal{L}_{\text{KD}}$. Dual-level distillation preserves the old *soft selection policy*: group preferences and patch-score distributions. It does not require the final hard top-B set to be identical.

Functional continuity: $\mathcal{L}_{\text{eq}}$. Selection identity should be allowed to change if an alternative evidence set is equally or more useful. The hinge

$$\mathcal{L}_{\text{eq}} = \max(0, U_{\text{old}} - U_{\text{new}})$$

therefore acts as a *floor, not a ceiling*: regression is penalized, improvement is free. This is the central difference between "remember the same evidence" and "retain the same evidence-selection competence."

Task competence: replay. $\mathcal{L}_{\text{diag}}^{\text{rep}}$ is simply the ordinary diagnostic loss on a replayed old slide. Replay is a data mechanism rather than a new mathematical objective. All within-task and continual terms update the *same current selector LoRA* in one backward pass:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{ev}} + \mathcal{L}_{\text{CL}}.$$

4. EXPERIMENTS

4.1. Protocol and measurements

We use the four-cohort TCGA benchmark introduced by ConSlide [11] in the form released by QPMIL-VL [13]: CLAM-tiled 256×256 patches at $10\times$, pre-extracted CONCH features, and ten patient-level folds. Tasks arrive in the reverse order ESCA→RCC→BRCA→LUNG, each a binary subtype task, for a final eight-class label space. Two protocols are used. The *comparison protocol* follows QPMIL-VL: ten folds, one run per fold with the seed equal to the fold index, and the metrics average accuracy (ACC), masked ACC (task-IL), forgetting [18], and backward transfer (BWT) [19]. The *ablation protocol* fixes fold 1 (2,273 training slides; test sizes 15/76/93/95) and varies five seeds with paired win counts. All runs use 5 epochs per task, learning rate 10^{-3}, $B = 8$, LoRA rank 4, β_s=β_u=0.1, unit λ's, and seeds 0–4. Every comparison was registered with a two-sided decision rule before the corresponding runs: win counts over the five seeds decide, the incumbent design is retained on a split, and one registered expectation that the data refuted is reported as measured (Sec. 4.3).

The controlled variants are sequential full fine-tuning of the selector (SeqFT), LoRA with boundary merging only, replay, replay+distillation, and full PathSelect (replay+distillation+utility hinge). Per-task specialists and joint offline training are references rather than deployable continual baselines.

We measure outcome and selection separately. Class-IL accuracy evaluates over all seen classes; task-IL accuracy restricts prediction to the task's own pair; leakage is the fraction predicted into another task's class set. Selection Jaccard, $|\mathcal{P}_{\text{own}} \cap \mathcal{P}_{\text{final}}|/|\mathcal{P}_{\text{own}} \cup \mathcal{P}_{\text{final}}|$, measures hard identity retention between the set selected immediately after learning a task and the set selected at the end. Jaccard is *evaluation only*; it is not the distillation loss. Functional retention is measured by

$$\Delta U = U_{\text{final}} - U_{\text{own}}.$$

With five seeds, comparisons are paired by seed and accompanied by win counts; 5/5 is called systematic, 4/5 directional, and no p-values are reported at this seed count.

4.2. Comparison with continual WSI methods

Table 1 places PathSelect in the continual WSI landscape under QPMIL-VL's splits and evaluation protocol. PathSelect reaches $0.854{\pm}.044$ ACC, $0.925{\pm}.019$ masked ACC, and $0.060{\pm}.041$ forgetting: the second-highest ACC among the continual methods in the table, 0.005 below QPMIL-VL's reported 0.859, and above every regularization, rehearsal, ConSlide, and prompt-based row; the offline joint-training reference reaches 0.908. ConSlide reports slightly lower forgetting (0.058 vs. 0.060) at substantially lower ACC (0.499), so we do not claim the lowest forgetting. PathSelect adapts 16.4k selector parameters per task, versus QPMIL-VL's reported 0.365M learnable parameters ($22.3\times$ fewer) and 0.5–39M for the MIL-based rows; these counts refer to different adapted components and are reported as adaptation-cost context, together with the absence of a prototype pool or task-specific module at inference and a 3.1 MB behavioral memory (a 30-WSI rehearsal buffer of features would occupy about 180 MB at 6.1 MB per slide).

4.3. What each part of PathSelect contributes

Learned selection is itself meaningful: in a preliminary budget sweep with a learned flat selector (inference only; seven budgets × four tasks), learned selection beat random selection in all 28 cells and peaked at $B = 8$ (0.8797 vs. 0.6607). Table 2 then removes one part of PathSelect at a time under the comparison protocol. The largest loss comes from removing preservation altogether: fine-tuning only the selector drops ACC from 0.830 to 0.452 and raises forgetting from 0.090 to 0.601, in every fold on every metric—while still beating full-model fine-tuning under the same splits (0.234 in Table 1), so the frozen pathway alone halves the collapse. Replay recovers most of the accuracy (0.821); distillation and the utility hinge then add 0.8 ACC points (7/10 folds) and remove 2.4 forgetting points (8/10); the hierarchy adds 2.5 ACC points (8/10) and removes 2.9 forgetting points (7/10) under the canonical reverse order. Under the forward order (QPMIL-VL's Table 1 protocol) the sign flips: the flat selector is better (0.841 vs. 0.811 ACC, 9/10 folds; Table 3), so the hierarchy's benefit is order-dependent; we retain it as the pre-registered choice on the canonical order and report both. Without any learned selection the frozen head reaches 0.679, so the policy contributes 17.5 points.

Table 4 separates the three preservation losses on the ablation protocol. Replay is the largest single contributor ($0.447 \rightarrow 0.778$) because it restores task attribution: with distillation alone, leakage stays at 0.323, three times the full method—in one seed 59 of 76 renal slides land in the lung classes—so distillation preserves *how* the selector scores but not *which task* the evidence belongs to. The utility hinge alone reaches 0.809; its increment on top of replay and distillation is order-sensitive and we report it as such. Under the hierarchy, removing the group level of distillation costs 3.95 task-IL and 3.71 class-IL points (5/5), so quota-level and patch-level preservation act at different resolutions.

Table 5 shows what accuracy cannot: unprotected selectors retain almost none of their original evidence (Jaccard 0.0023/0.0010) and lose old-task utility by an order of magnitude more than protected ones (ΔU $-220/-301$ vs. -16.8). LoRA merging alone shows no systematic difference from full fine-tuning (3/5): merging is the deployment substrate, not the preservation mechanism. Distillation without the hinge retains slightly more identity (0.1752 vs. 0.1613) yet is less accurate and forgets more: preserving *where* the policy looks does not secure *how useful* what it finds is—the gap the utility hinge closes. Three further interventions raised Jaccard without improving diagnosis: warm-starting task 1 with full-parameter training, damping each fold-in to half, and composing independently trained residuals post hoc.

Update-protocol probes. Each was decided by a rule fixed before the run. Warm start and damped merging showed no systematic benefit. A single adapter never reset lost 3.54 class-IL points on the flat substrate (4/5) and became a trade-off under the hierarchy (systematic task-IL edge for fresh adapters, systematically lower leakage for the single adapter); fresh adapters are retained. Post-hoc composition refuted our registered expectation: independently trained residuals, summed once, beat the bare sequential substrate on every metric in every seed, consistent with strong interference in unprotected sequential updates (cf. task arithmetic [28])—yet trailed the

Table 1. **Comparison under QPMIL-VL's reverse-order splits and evaluation protocol** (ESCA→RCC→BRCA→NSCLC, ten-fold CV, mean±sd). Published rows are quoted from QPMIL-VL's Table 2 [13] and were not rerun by us; PathSelect uses the identical public split files and metric definitions, while training implementations and hyperparameters follow each method and are therefore not controlled across papers. Unlike the cited MIL baselines, PathSelect adapts only the evidence selector and keeps the diagnostic pathway frozen, so parameter counts characterize adaptation cost rather than capacity-matched architectures; external rows provide published benchmark context rather than a controlled reimplementation comparison. Buffer sizes in WSIs; PathSelect's Selection Memory holds 512 behavioral snapshots (3.1 MB, excluding the feature store required for replay). Forward-order results are in Table 3.

Type	Method	ACC↑	Forgetting↓	BWT↑	Masked ACC↑	Buffer
Published methods—values quoted from [13]						
Reference	JointTrain (offline reference)	0.908±.022	—	—	0.937±.022	—
Reference	FineTune (lower)	0.234±.008	0.927±.023	−0.927±.023	0.803±.060	0
Regularization	EWC [20]	0.235±.011	0.928±.020	−0.928±.020	0.833±.069	0
Regularization	LwF [21]	0.236±.016	0.908±.030	−0.908±.030	0.900±.041	0
Rehearsal	A-GEM [22]	0.536±.047	0.527±.067	−0.527±.067	0.872±.026	30
Rehearsal	ER-ACE [23]	0.703±.049	0.281±.062	−0.279±.063	0.889±.041	30
Rehearsal	DER++ [24]	0.684±.055	0.310±.069	−0.307±.072	0.910±.043	30
Rehearsal	ER [25]	0.644±.028	0.387±.050	−0.386±.049	0.901±.035	30
WSI CL	ConSlide [11]	0.499±.025	0.058±.032	−0.021±.039	0.854±.039	30
VLM	AttriCLIP [26]	0.694±.058	0.207±.063	−0.207±.063	0.861±.019	0
VLM	MI-Zero [27]	0.839±.034	—	—	0.909±.019	0
VLM WSI CL	QPMIL-VL [13]	0.859±.032	0.064±.031	−0.064±.031	0.925±.018	0
Our method—run under the same splits and metrics						
Selection CL	PathSelect (ours)	0.854±.044	0.060±.041	−0.056±.038	0.925±.019	512 snap. (3.1 MB)

Table 2. **Leave-out ablation under the comparison protocol** (ten-fold CV, mean±sd). Parentheses: folds in which the configuration one row above is better, paired by fold.

Configuration	ACC↑	Forgetting↓	Masked ACC↑
Full PathSelect	0.854±.044	0.060±.041	0.925±.019
− hierarchy (flat selector)	0.830±.029 (8/10)	0.090±.043 (7/10)	0.914±.021 (8/10)
− distillation, − utility hinge (replay only)	0.821±.036 (7/10)	0.114±.038 (8/10)	0.919±.026 (4/10)
− all preservation (selector fine-tuning)	0.452±.071 (10/10)	0.601±.122 (10/10)	0.811±.054 (10/10)
− learned selection (frozen head, mean pooling)	0.679±.032	—	0.776±.029

Table 3. **Order robustness** (ten-fold CV ACC, mean±sd). QPMIL-VL values are reported in [13] (Tables 1–2). The hierarchical selector is better in the reverse order (8/10 folds) and worse in the forward order (9/10 folds).

Task order	PathSelect (hier.)	PathSelect (flat)	QPMIL-VL (reported)
ESCA→RCC→BRCA→LUNG (reverse)	0.854±.044	0.830±.029	0.859±.032
LUNG→BRCA→RCC→ESCA (forward)	0.811±.029	0.841±.043	0.890±.021

full method by 13.1 points (5/5), because independence excludes replay and distillation targets require an accumulated model; sequential consolidation earns its place only together with preservation. Porting MergeSlide's orthogonal-projection merging [16] to our selector did not close this gap (0.709±.035; no systematic difference from plain summation, 3/5; the released implementation's diagonal mask is a no-op and the two variants differ within noise). With 128 memory entries PathSelect still outperformed replay with 512 (task-IL +3.06, 5/5), reported for task-IL and this range only. A task-conditioned query, a stateful multi-round selector, and a group-level semantic prior failed their registered gates and were removed.

Table 4. **Preservation-component ablation** (flat, fold 1, 5-seed means). Replay restores task attribution, the utility hinge assigns credit, distillation stabilizes scoring; no single term or pair matches the full set.

Replay	Distill.	Utility	Class-IL↑	Task-IL↑	Leak.↓
			.447	.811	.476
	✓		.600	.853	.323
		✓	.809	.885	.091
✓			.778	.907	.104
✓	✓		.797	.897	.117
✓	✓	✓	**.824**	**.915**	**.101**

Table 5. **Selection-level metrics under the ablation protocol** (flat, fold 1, mean over 5 seeds): what accuracy alone cannot show.

Method	Class-IL↑	Sel. Jaccard↑	Leakage↓	ΔU ↑
Sequential fine-tuning	0.4774	0.0023	0.4408	−220
LoRA merge only	0.4466	0.0010	0.4756	−301
+ replay	0.7778	0.0864	0.104	−21.2
+ distillation	0.7972	0.1752	0.117	−31.9
PathSelect (full)	0.8239	0.1613	0.1005	−16.8

5. DISCUSSION AND LIMITATIONS

PathSelect suggests that continual learning in budgeted perception should distinguish two failure modes that accuracy alone conflates. A policy may change *where* it looks, and the new evidence may or may not remain *functionally equivalent*. This is why Jaccard, KD, utility, and the utility hinge should not be collapsed into one notion. KD preserves a soft behavior distribution; Jaccard measures hard identity after the fact. Candidate utility u_i teaches which patch is useful now; set utility U evaluates whether the selected evidence remains useful across time. The utility-preservation hinge uses U as a lower bound, deliberately allowing better alternative evidence.

The study also reveals what is *not* sufficient. LoRA merging alone does not protect the selector. Warm start, damped merging, and post-hoc composition each increase selection Jaccard without improving diagnosis (Sec. 4.3), so "choose the same patches" is not a complete CL objective: selection stability is not sufficient for diagnostic preservation, and identity and function must be measured separately. This motivates behavioral and functional preservation as complementary constraints rather than redundant regularizers.

Our claims remain bounded. The benchmark uses four organ-separated TCGA tasks, one frozen foundation model (CONCH), one main budget ($B = 8$), known task boundaries during training, and five seeds. The tasks are highly separable (a linear probe separates them at 98.21%), and explicit task conditioning failed its registered gate, so harder within-organ or mixed-task streams may yield different conclusions. PathSelect is also a one-shot evidence-acquisition policy, not a multi-step navigation agent. Its task-free shared selector can serve as a foundation for continual evidence-navigation systems, but sequential stateful acquisition is left to future work. The memory is compact but is not claimed to provide a formal privacy guarantee. The hierarchical substrate's advantage is order-dependent (Table 3); the preservation mechanism was pre-registered and validated on the canonical reverse order, and its forward-order counterpart is reported as measured.

6. CONCLUSION

PathSelect reframes continual WSI learning around a simple but under-measured question: after learning new diagnostic tasks, does the model still know *where to look*? By freezing the diagnostic pathway, we isolate the evidence selector as the study object. Within tasks, semantic and counterfactual signals teach the selector to find useful evidence. Across tasks, a single merged selector is preserved at two complementary levels: distillation maintains selection behavior, while a utility-preservation hinge allows evidence identity to change without allowing usefulness to regress; replay maintains old-task competence. The resulting method substantially reduces forgetting at zero inference-time memory or routing cost. More broadly, the results argue that continual learning for budgeted perception should preserve not only what a model predicts, but also the policy by which it acquires the evidence used to predict.

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